

MediQueue AI Healthcare Management System

End-to-End Demo Flow

1. Login as Admin

Login using Admin credentials.

Dashboard displays:

- Total Patients
 - Total Doctors
 - Total Appointments
 - Queue Status
-

2. Search Existing Patient

Search Mobile Number:

9876543210

Result:

- Patient Name: Archana Patel
 - Mobile Number: 9876543210
-

3. Book Appointment

Create Appointment:

- Patient: Archana Patel
- Doctor: Dr. Priya Mehta (ENT)
- Date: Tomorrow 10:00 AM

Result:

- Appointment Created Successfully
 - Queue Number Generated: 12
 - Status: PENDING
-

4. View Appointment List

Patient	Doctor	Appointment Date	Status
Archana Patel	Dr. Priya Mehta	13 Jun 2026 10:00 AM	PENDING

5. Queue Management

Queue Screen:

- Queue No: 12
- Patient: Archana Patel
- Status: CONFIRMED

Action:

Call Patient

Result:

Status changes to:

IN_PROGRESS

6. Complete Consultation

Doctor Notes:

Patient has ear infection. Prescribed antibiotics for 5 days.

Action:

Complete Appointment

Result:

- Status → COMPLETED
- Doctor Notes Saved
- AI Summary Generated

Example AI Summary:

Patient diagnosed with mild ear infection and prescribed antibiotics for five days. Follow-up recommended if symptoms continue.

7. Patient History

View:

- Previous Appointments
 - Consultation Notes
 - AI Summaries
 - Uploaded Reports
 - Prescriptions
-

8. AI Medical Report Analyzer

Upload:

- Blood Test Report
- Lab Report PDF

AI Analysis:

- Key Findings
- Abnormal Values
- Simplified Explanation
- AI Generated Summary

Example:

Hemoglobin: Low

Vitamin D: Deficient

AI Summary:

Patient shows signs of mild anemia and Vitamin D deficiency. Physician consultation is recommended.

9. AI Hospital Assistant (RAG)

User:

What are hospital timings?

AI Response:

MediQueue Hospital operates from 9:00 AM to 8:00 PM Monday through Saturday.

Emergency services are available 24 hours.

10. AI Symptom Analysis

User:

I have ear pain and hearing issues.

AI Response:

You should consult an ENT specialist.

Recommended Doctors:

- Dr. Priya Mehta
 - Dr. Amit Joshi
-

11. AI Appointment Booking

User:

Book appointment with Dr. Priya Mehta tomorrow at 10 AM

AI Response:

Appointment booked successfully.

Doctor: Dr. Priya Mehta

Queue Number: 13

Status: PENDING

12. Emergency Information

User:

What is the emergency contact number?

AI Response:

Emergency Helpline:

+91-9999999999

Available 24 Hours

AI Scheduling Features

Smart Time Slot Management

Supported Examples:

- Book appointment tomorrow
 - Book appointment tomorrow at 2 PM
 - Book appointment tomorrow at 10:30 AM
 - Book appointment next Monday
 - Book appointment on June 25
-

Conflict Detection

Input:

Book appointment with Dr Patel tomorrow at 2 PM

Result:

Doctor already has an appointment at 14:00.

Please choose another time.

Alternative Slot Suggestions

Result:

Doctor already has an appointment at 14:00.

Available slots:

✓ 10:00

✓ 11:00

✓ 12:00

Reply with a preferred time.

Working Hours Validation

Input:

Book appointment tomorrow at 8 PM

Result:

Doctor appointments are available only between:

09:00 AM – 06:00 PM

Lunch Break Protection

Input:

Book appointment tomorrow at 1 PM

Result:

Doctor is unavailable at 13:00.

Lunch Break:

01:00 PM – 02:00 PM

Daily Capacity Control

Condition:

20 appointments already booked.

Result:

Doctor is fully booked for this date.

Please choose another day.

Urgent Symptom Detection

Input:

I have chest pain and difficulty breathing.

Result:

 Urgent Symptoms Detected

Please seek immediate medical attention.

System Test Cases

Test Case	Description	Result
TC-01	Book appointment tomorrow	Passed
TC-02	Book appointment tomorrow at 2 PM	Passed
TC-03	Book appointment at 10:30 AM	Passed
TC-04	Missing doctor name validation	Passed
TC-05	Doctor not found validation	Passed
TC-06	Past date validation	Passed
TC-07	Conflict detection	Passed
TC-08	Alternative slot suggestion	Passed
TC-09	Working hours validation	Passed
TC-10	Lunch break validation	Passed
TC-11	Daily capacity limit	Passed
TC-12	Symptom analysis	Passed
TC-13	Emergency detection	Passed
TC-14	AI report analyzer	Passed
TC-15	RAG hospital assistant	Passed
TC-16	Queue management workflow	Passed
TC-17	Consultation completion workflow	Passed
TC-18	AI appointment booking	Passed

Test Case	Description	Result
TC-19	Patient history tracking	Passed
TC-20	Emergency information retrieval	Passed

Test Summary

Total Test Cases: 20

Passed: 20

Failed: 0

Success Rate: 100%

Key Highlights

- ✓ Role-Based Access Control
- ✓ Patient Management
- ✓ Doctor Management
- ✓ Appointment Scheduling
- ✓ Queue Management
- ✓ Medical History Tracking
- ✓ AI Medical Report Analyzer
- ✓ AI-Powered Hospital Assistant
- ✓ RAG Knowledge Base
- ✓ Symptom-Based Doctor Recommendation
- ✓ AI Appointment Booking
- ✓ AI Scheduling Engine

- ✓ Alternative Slot Suggestions
- ✓ Working Hours Validation
- ✓ Conflict Detection
- ✓ Daily Capacity Management
- ✓ Urgent Symptom Detection
- ✓ AI Consultation Summary
- ✓ Emergency Assistance
- ✓ Production-Ready Healthcare MVP